



The CPRI (RSOP) Sponsored Project to RKDF University, Bhopal

**“Post Combustion Carbon Capture & Sequestration (CCS)
Plant on a Coal Fired Power Plant - Feasibility Study”**

Sanctioned vide: CPRI/R&D/TC/Thermal/ 2019 Dated 06-02-2019

Period of Project: April 2019 – June 2021

Collaborator: RGPV, Bhopal & RPI, USA

PROJECT COMPLETION REPORT

TECHNICAL REPORT

SUBMITTED BY: DR. VINOD KRISHNA SETHI, DG (Research) RKDF University

Principal Investigator

**Executing Organization: Ram Krishna Dharmarth Foundation (RKDF)
University, Airport Bypass Road, Gandhi Nagar Bhopal, MP 462033**



RKDF UNIVERSITY

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TECHNICAL REPORT

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CHAPTER – 1

INTRODUCTION

CHAPTER – 1.1

PREAMBLE

CHAPTER – 1 INTRODUCTION

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1.1.1 Genesis of the Study

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1.1 Preamble – The Project & the Technical Report

A. THE PROJECT: Sponsored by CPRI under RSOP scheme vide Sanction letter no. CPRI/R&D/ TC/ Thermal / 2019 Dated 06-02-2019 (Period: April 2019 to June 2021)

Project Title:

“Post Combustion Carbon Capture & Sequestration (CCS) Plant on a Coal Fired Power Plant - Feasibility Study”

Total Outlay:

- CPRI Funding: Rs. 35.00 Lacs
- CPRI Monitoring : Rs. 3.50 Lacs (To CPRI Thermal R&D Division)
- RKDF Equity: Rs. 15.00 Lacs: For Scheffler Discs of Solar Thermal , a Coal fired Steam Boiler , Foundations , Manpower, Testing Commissioning & Contingency

PI: Dr. Vinod Krishna Sethi, DG (Research), RKDF University, Ex. VC RKDF & Ex. Pro-VC RGPV & Ex. Director MOP/ CEA

CO- PIs: Dr. Sunil K Gupta, VC, RGPV – State Tech. University of MP State & **Dr. Savita Vyas**, Professor, RGPV

EPC Contractor: M/S SUNRISE CSP (I) Private Ltd. Baroda

Executing Organization: Ram Krishna Dharmarth Foundation (RKDF) University, Airport Bypass Road, Gandhi Nagar Bhopal, MP 462033

Executives of RKDF University:

- **Dr. Sunil Kapoor**, Hon’ble Chairman RKDF Group
- **Dr. Sadhna Kapoor**, Hon’ble Chancellor
- **Dr. Sudesh K Sohani**, Hon’ble VC
- **Mr. Siddharth Kapoor**, MD,
- **Dr. B N Singh**, DG(Management)
- **Dr. Sunil Patil**, Director
- **Mr. Suresh K Rathore**, Controller of Finance
- **Dr. Ravi Kumar Singh Pippal**, PDF Fellow & Professor

Project Location: Sir J C Bose Interdisciplinary Technology Park, RKDF, University, Bhopal

B. THE TECHNICAL REPORT

Chapter- wise summary of each of the Chapters of this ‘Technical Report’ is given here under:

CHAPTER – 1: INTRODUCTION

In addition to the Preamble, this chapter covers the ‘Genesis of present R&D Project’ and salient features of the pilot project covering specification analysis of test results and a one page ‘Executive summary’.

Detailed Literature survey giving ‘Gaps’ in the understanding of the Research subject, International Perspective, State of Art of Technology, Policy aspects, Opportunities and Economic implications of CCS Projects in Indian Context are given in the sub-sections of this chapter. Research Papers on Solar Integrated CCS are given due emphasis in this Chapter.

Full text of the ‘References & Papers’ selected for study are given at **ENCL: 01** for information.

Chapter- 2 & 3: OBJECTIVES, SCOPE OF THE STUDY & DELIVERABLES

These Chapters give a summary table of how the Objectives of the Sponsored project have been achieved through various out-come results and deliverables.

Chapter - 4: EXPERIMENTAL PROCEDURE

The chapter covers the Specifications and Pilot Plant Design parameters, its integration with Solar Thermal for regeneration of Solvent and details of Coal fired boiler for Steam & Flue Gas generation. Thirty days continuous testing trial Run Data Sheet also forms the part of this chapter.

Chapter – 5: RESULTS, DISCUSSION & ANALYSIS – PRIME OUTPUT THE STUDY

This Chapter covers analysis of out-put of study and various Sequestration Options like; Production of Hydrogen, Pilot study of production of Bio-diesel through Algae Route and sequestration options like use of Captured CO₂ to depleted Coal mines etc.

A SCADA / DATA Acquisition System (DAS) is under installation but its commissioning is under hold due to lock-down and severe Pandemic condition. As such full report on this will be submitted as the situation improves as a ‘Supplementary Report’. As such, the Energy Penalty optimization in regeneration of Solvent through Steam from Solar Thermal Plant has been determined and results presented through simulation software **ASPEN Plus**.

Chapter -6: CONCLUSIONS

Main Conclusions of the R&D Pilot Project together with Future Perspective covering Sustainable development aspects and Issues in implementation in Industry & Academia are presented

Chapter -7: TECHNOLOGY KNOW-HOW REPORT

This chapter is devoted to Scaling up of CO₂ Capture Plant on a 500 MW Coal fired Thermal Unit at ANPARA B TPS -2x500 MW Unit no. 4 of 500 MW. The water, Power and Land requirement as well as project authority's perspective towards deployment of this technology in their power plants are presented. The Heat / Energy Balances, HBD/ PI diagrams of the 500 MW Unit at ANPARA TPS are also presented.

Chapter -8: PROGRESS REPORTS, SOE & UC SUBMITTED TO CPRI

A summary of latest progress report submitted to CPRI has been presented here and full details will be found in the "Folder" on Progress Reports / SOE / UC (**ENCL: 02**)

Chapter - 9: PUBLICATIONS & PATENTS

A compiled list of publications and book chapters emerged through the CPRI sponsored RSOP project has been given in this chapter, while the full text of these publications will be found in **ENCL: 03**. The patent application entitled "A reactor for reducing energy penalty in a solar thermal integrated Carbon capture plant with 24x7 energy storage" has been submitted as detailed in **ENCL: 03**.

Chapter - 10: PFMS SUMMARY & EQUITY

This Chapter is devoted to Summary of Payments through PFMS of the CPRI Grant and Equity participation by RKDF University on Civil works, procurement of Coal fired Boiler & Solar Thermal Plant and Testing & Commissioning expenses.